



Linux Fundamentals

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TABLE OF CONTENT

FILE OPERATION	1
Create and Navigate to a Lab Directory:	1
Create and Edit a File:	1
View File Content:	1
Copy and Rename:	1
Remove File and Directories:	2
Check Permission:	2
APT PACKAGE MANAGEMENT	3
Update the Package List:	3
Install the C Compiler:	3
Verify Installation:	3
Search for other packages:	3
C COMPILATION:	4
Create a C source File:	4
Compile and Run:	5
Modify and Recompile:	5
CLEANUP AND VERIFICATION	6
Go back to Home Directory	6
List all files in the Lab Directory with Details	6
View the Content of C File	6
Remove the Entire Lab Directory	6

LIST OF FIGURES

Figure 01: Lab Directory Navigation inside os-labs Folder	1
Figure 02: greeting.txt Nano Editor	1
Figure 03: greeting.txt Content View	1
Figure 04: Copy greeting.txt to backup.txt	1
Figure 05: Move backup.txt to old_greeting.txt	2
Figure 06: Remove old_greeting.txt	2
Figure 07: Create and Delete temp_dir	2
Figure 08: Permission Check in greeting.txt	2
Figure 09: Package List Update with apt	3
Figure 10: build-essential Installation	3
Figure 11: GCC Install Verification	3
Figure 12: Package Searching	4
Figure 13: hello.c Source Code in Nano Editor	4
Figure 14: hello.c Source Code View with cat	4
Figure 15: hello.c Compilation and Output	5
Figure 16: -Wall Flag Compilation and Output	5
Figure 17: Modified hello.c Source Code in Nano Editor	5
Figure 18: Recompiling and Running Updated hello_program	5
Figure 19: Home Directory	6
Figure 20: Detailed File Listing in linux_lab Directory	6
Figure 21: Final Content Check of hello.c	6
Figure 22: Removing the linux_lab Directory	6

FILE OPERATION

Create and Navigate to a Lab Directory:

```
Microsoft Windows [Version 10.0.26200.9278]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>wsl -d Ubuntu --cd ~/os-labs
wagleayden@Ayden:~/os-labs$ ls
week1
wagleayden@Ayden:~/os-labs$ cd week1
wagleayden@Ayden:~/os-labs/week1$ ls
practiceayd practiceayd.c
wagleayden@Ayden:~/os-labs/week1$ sudo mkdir linux_lab
[sudo: authenticate] Password:
wagleayden@Ayden:~/os-labs/week1$ ls
linux_lab practiceayd practiceayd.c
wagleayden@Ayden:~/os-labs/week1$ cd linux_lab
```

Figure 01: Lab Directory Navigation inside os-labs Folder

In the terminal, there is a folder opened named 'os-lab'. Inside week1, a working directory is created using the **sudo mkdir command**. This directory served as the main workspace for future operations.

Create and Edit a File:

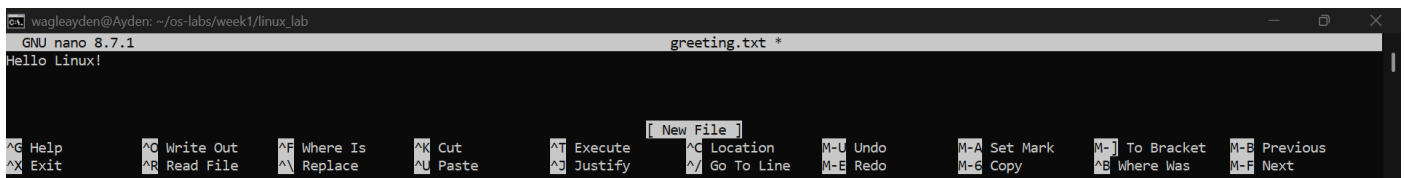
The image shows a terminal window with the Nano text editor open. The editor's title bar indicates it is editing 'greeting.txt'. The content of the file is 'Hello Linux!'. The Nano editor's menu bar is visible at the bottom, showing various commands like Help, Exit, Write Out, Read File, Where Is, Replace, Cut, Paste, Execute, Justify, Location, Go To Line, Undo, Redo, Set Mark, Copy, To Bracket, Where Was, Previous, and Next.

Figure 02: greeting.txt Nano Editor

A file named 'greeting.txt' was created using **sudo nano greeting.txt** where 'Hello, Linux!' text was added. Then it is saved using Ctrl + X, Y for saving modified change and Enter.

View File Content:

```
wagleayden@Ayden:~/os-labs/week1/linux_lab$ sudo nano greeting.txt
wagleayden@Ayden:~/os-labs/week1/linux_lab$ cat greeting.txt
Hello linux!
wagleayden@Ayden:~/os-labs/week1/linux_lab$
```

Figure 03: greeting.txt Content View

To view the greeting.txt file without opening the folder, **cat** command was used. The command displayed the content written in nano editor.

Copy and Rename:

```
wagleayden@Ayden:~/os-labs/week1/linux_lab$ ls
greeting.txt
wagleayden@Ayden:~/os-labs/week1/linux_lab$ sudo cp greeting.txt backup.txt
wagleayden@Ayden:~/os-labs/week1/linux_lab$ ls
backup.txt greeting.txt
wagleayden@Ayden:~/os-labs/week1/linux_lab$
```

Figure 04: Copy greeting.txt to backup.txt

Then the greeting.txt was copied into a new text file named 'backup.txt.' It was done using **cp** command. When the files are listed again, there is a new text file.

```
wagleayden@Ayden:~/os-labs/week1/linux_lab$ ls
backup.txt  greeting.txt
wagleayden@Ayden:~/os-labs/week1/linux_lab$ sudo mv backup.txt old_greeting.txt
wagleayden@Ayden:~/os-labs/week1/linux_lab$ ls
greeting.txt  old_greeting.txt
wagleayden@Ayden:~/os-labs/week1/linux_lab$
```

Figure 05: Move backup.txt to old_greeting.txt

To move the file to a new text file, **mv** command was used. The backup.txt was replaced with old_greeting.txt.

Remove File and Directories:

```
wagleayden@Ayden:~/os-labs/week1/linux_lab$ ls
greeting.txt  old_greeting.txt
wagleayden@Ayden:~/os-labs/week1/linux_lab$ sudo rm old_greeting.txt
wagleayden@Ayden:~/os-labs/week1/linux_lab$ ls
greeting.txt
wagleayden@Ayden:~/os-labs/week1/linux_lab$
```

Figure 06: Remove old_greeting.txt

```
wagleayden@Ayden:~/os-labs/week1/linux_lab$ sudo mkdir temp_dir
wagleayden@Ayden:~/os-labs/week1/linux_lab$ ls
greeting.txt  temp_dir
wagleayden@Ayden:~/os-labs/week1/linux_lab$ sudo rmdir temp_dir
wagleayden@Ayden:~/os-labs/week1/linux_lab$ ls
greeting.txt
wagleayden@Ayden:~/os-labs/week1/linux_lab$
```

Figure 07: Create and Delete temp_dir

To remove the old_greeting.txt, **rm** command was used. There is another command called **rm-r** that is used to forcefully delete any folder or file. The result shows the files and folder being deleted.

Check Permission:

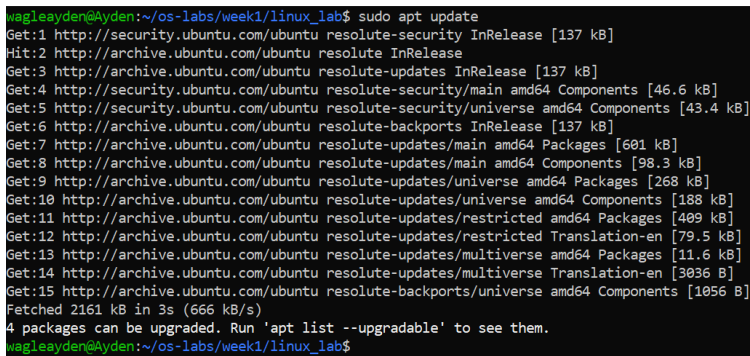
```
wagleayden@Ayden:~/os-labs/week1/linux_lab$ ls -l greeting.txt
-rw-r--r-- 1 root root 13 Sep  8 05:54 greeting.txt
wagleayden@Ayden:~/os-labs/week1/linux_lab$
```

Figure 08: Permission Check in greeting.txt

To inspect its file permissions and metadata, **ls -l** command was used. The command showed **-rw-r--r--**. The file had read and write permission for the owner but read only permission for groups and others. The size was 13 bytes, It was last modified on 8th September at 05:54.

APT PACKAGE MANAGEMENT

Update the Package List:

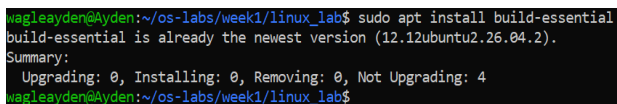


```
wagleayden@Ayden:~/os-labs/week1/linux_lab$ sudo apt update
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Hit:2 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.6 kB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.4 kB]
Get:6 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [601 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [98.3 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [268 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [188 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [409 kB]
Get:12 http://archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [79.5 kB]
Get:13 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 Packages [11.6 kB]
Get:14 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse Translation-en [3036 B]
Get:15 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Components [1056 B]
Fetched 2161 kB in 3s (666 kB/s)
4 packages can be upgraded. Run 'apt list --upgradable' to see them.
wagleayden@Ayden:~/os-labs/week1/linux_lab$
```

Figure 09: Package List Update with apt

To refresh the local packages from Ubuntu, **sudo apt update** command was used. It fetched the repositories like security, main, universe, etc. It also indicated that 4 packages could be upgraded.

Install the C Compiler:

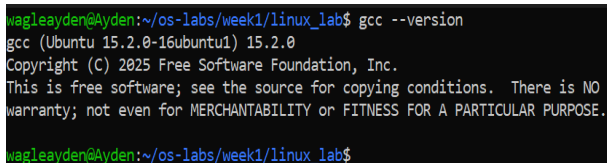


```
wagleayden@Ayden:~/os-labs/week1/linux_lab$ sudo apt install build-essential
build-essential is already the newest version (12.12ubuntu2.26.04.2).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 4
wagleayden@Ayden:~/os-labs/week1/linux_lab$
```

Figure 10: build-essential Installation

To install the essential packages for compiling C programs, **sudo apt install build-essential** command was used. These packages are required for GCC compilers. The output shows that it was already installed with the newest version.

Verify Installation:

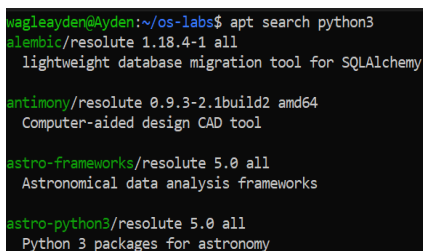


```
wagleayden@Ayden:~/os-labs/week1/linux_lab$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
wagleayden@Ayden:~/os-labs/week1/linux_lab$
```

Figure 11: GCC Install Verification

To confirm that GNU C Compiler was installed, **gcc - - version** was used. The output displayed there is GCC version 15.2.0 with copyright and license information.

Search for other packages:



```
wagleayden@Ayden:~/os-labs$ apt search python3
alembic/resolute 1.18.4-1 all
  lightweight database migration tool for SQLAlchemy

antimony/resolute 0.9.3-2.1build2 amd64
  Computer-aided design CAD tool

astro-frameworks/resolute 5.0 all
  Astronomical data analysis frameworks

astro-python3/resolute 5.0 all
  Python 3 packages for astronomy
```

```

bamkit/resolute 0.0.1+git20170413.ccd079d-3build1 all
tools for common BAM file manipulations

beancount/resolute 3.2.0-2build1 all
Double-entry accounting from text files

berrynet-dashboard/resolute 3.10.2-2build1 all
deep learning gateway - python3 modules

brlTTY/resolute 6.7-1ubuntu6 amd64
Access software for a blind person using a braille display

budgie-app-launcher-applet/resolute-updates 2.2.3-1ubuntu1.1 amd64
Applet to provide an alternative means to launch applications

budgie-brightness-controller-applet/resolute-updates 2.2.3-1ubuntu1.1 amd64
Applet to control the brightness of the screen

budgie-clockworks-applet/resolute-updates 2.2.3-1ubuntu1.1 amd64
Applet to display clock across multiple time zones

budgie-countdown-applet/resolute-updates 2.2.3-1ubuntu1.1 amd64
Applet providing a countdown capability on the Budgie Desktop

budgie-dropby-applet/resolute-updates 2.2.3-1ubuntu1.1 amd64
Applet to popup when a USB device is connected

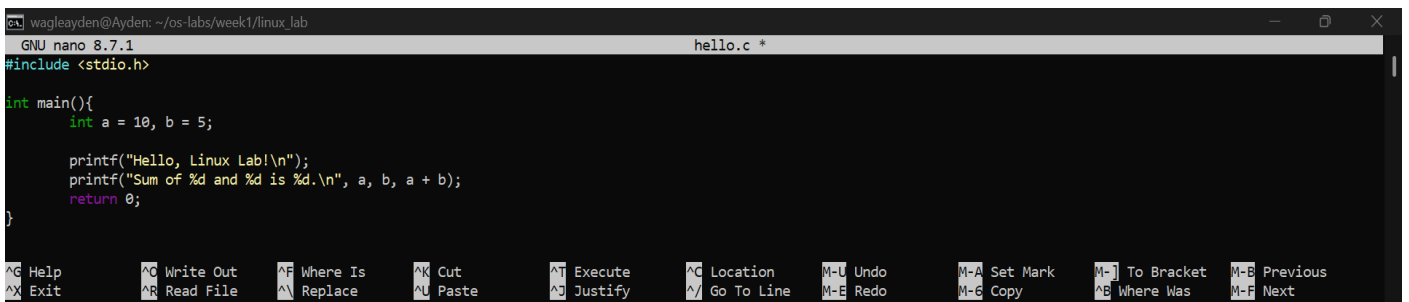
```

Figure 12: Package Searching

To check the available package related to Python 3, **apt search python3** was used. The output returned a matching package. It is to demonstrate the use of **apt search** to find other software packages too.

C COMPILATION:

Create a C source File:



```

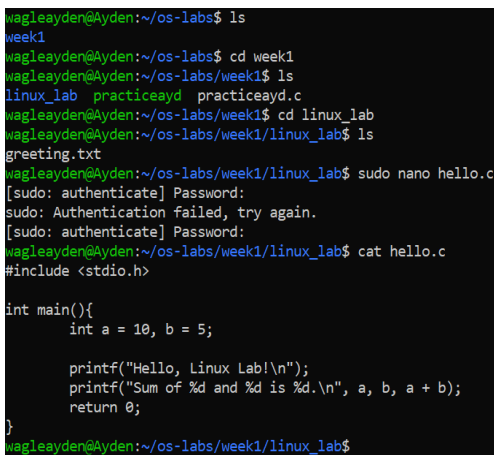
GNU nano 8.7.1 hello.c *
#include <stdio.h>

int main(){
    int a = 10, b = 5;

    printf("Hello, Linux Lab!\n");
    printf("Sum of %d and %d is %d.\n", a, b, a + b);
    return 0;
}

```

Figure 13: hello.c Source Code in Nano Editor



```

wagleayden@Ayden:~/os-labs$ ls
week1
wagleayden@Ayden:~/os-labs$ cd week1
wagleayden@Ayden:~/os-labs/week1$ ls
linux_lab  practiceayd  practiceayd.c
wagleayden@Ayden:~/os-labs/week1$ cd linux_lab
wagleayden@Ayden:~/os-labs/week1/linux_lab$ ls
greeting.txt
wagleayden@Ayden:~/os-labs/week1/linux_lab$ sudo nano hello.c
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
wagleayden@Ayden:~/os-labs/week1/linux_lab$ cat hello.c
#include <stdio.h>

int main(){
    int a = 10, b = 5;

    printf("Hello, Linux Lab!\n");
    printf("Sum of %d and %d is %d.\n", a, b, a + b);
    return 0;
}
wagleayden@Ayden:~/os-labs/week1/linux_lab$

```

Figure 14: hello.c Source Code View with cat

A C source file named hello.c was created using the Nano text editor. The program was a basic C program to add a and b with values 10 and 5. Then we checked the content with **cat** command.

Compile and Run:

```
wagleayden@Ayden:~/os-labs/week1/linux_lab$ sudo gcc hello.c -o hello_program
wagleayden@Ayden:~/os-labs/week1/linux_lab$ ./hello_program
Hello, Linux Lab!
Sum of 10 and 5 is 15.
wagleayden@Ayden:~/os-labs/week1/linux_lab$
```

Figure 15: *hello.c* Compilation and Output

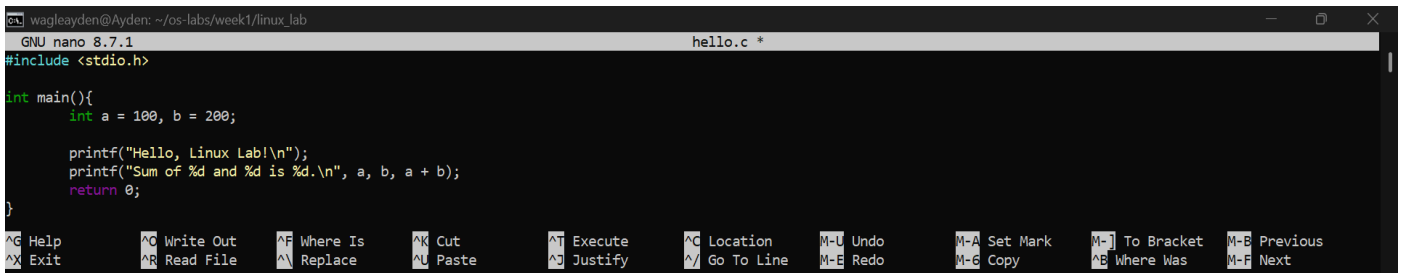
```
wagleayden@Ayden:~/os-labs/week1/linux_lab$ sudo gcc -Wall hello.c -o hello_program_warn
wagleayden@Ayden:~/os-labs/week1/linux_lab$ ./hello_program_warn
Hello, Linux Lab!
Sum of 10 and 5 is 15.
wagleayden@Ayden:~/os-labs/week1/linux_lab$
```

Figure 16: *-Wall* Flag Compilation and Output

The `hello.c` file was compiled using the `gcc` compiler with the command `sudo gcc hello.c -o hello_program`. When the code is executed, the program displays the addition of two integers.

Then the source file is compiled again using the `-Wall` flag. The command was `sudo gcc -Wall hello.c -o hello_program_warn`. It stands for ‘*all warning*’ which makes the code compile unless there is a hard error. Its main purpose is to help catch potential bugs, bad practices, or mistakes in code before they cause problems at runtime.

Modify and Recompile:



```
GNU nano 8.7.1 hello.c *
#include <stdio.h>

int main(){
    int a = 100, b = 200;

    printf("Hello, Linux Lab!\n");
    printf("Sum of %d and %d is %d.\n", a, b, a + b);
    return 0;
}
```

Figure 17: Modified *hello.c* Source Code in Nano Editor

```
wagleayden@Ayden:~/os-labs/week1/linux_lab$ sudo nano hello.c
wagleayden@Ayden:~/os-labs/week1/linux_lab$ cat hello.c
#include <stdio.h>

int main(){
    int a = 100, b = 200;

    printf("Hello, Linux Lab!\n");
    printf("Sum of %d and %d is %d.\n", a, b, a + b);
    return 0;
}
wagleayden@Ayden:~/os-labs/week1/linux_lab$ sudo gcc -Wall hello.c -o hello_program
wagleayden@Ayden:~/os-labs/week1/linux_lab$ ./hello_program
Hello, Linux Lab!
Sum of 100 and 200 is 300.
wagleayden@Ayden:~/os-labs/week1/linux_lab$
```

Figure 18: *Recompiling and Running Updated hello_program*

The `hello.c` file was reopened in the Nano editor. The integers values were changed to 100 and 200. Then it was saved and recompiled.

CLEANUP AND VERIFICATION

Go back to Home Directory

```
wagleayden@Ayden:~/os-labs/week1/linux_lab$ ls
greeting.txt  hello.c  hello_program  hello_program_warn
wagleayden@Ayden:~/os-labs/week1/linux_lab$ cd ~
wagleayden@Ayden:~$
```

Figure 19: Home Directory

To navigate back to the home directory, **cd ~** command was used.

List all files in the Lab Directory with Details

```
wagleayden@Ayden:~/os-labs/week1/linux_lab$ ls
greeting.txt  hello.c  hello_program  hello_program_warn
wagleayden@Ayden:~/os-labs/week1/linux_lab$ cd ~
wagleayden@Ayden:~$ ls -lh ~/os-labs/week1/linux_lab/
total 40K
-rw-r--r-- 1 root root 13 Sep  8 05:54 greeting.txt
-rw-r--r-- 1 root root 152 Sep  8 06:30 hello.c
-rwxr-xr-x 1 root root 16K Sep  8 06:31 hello_program
-rwxr-xr-x 1 root root 16K Sep  8 06:26 hello_program_warn
wagleayden@Ayden:~$
```

Figure 20: Detailed File Listing in linux_lab Directory

To display a detailed listing of all files within the lab directory, **ls -lh ~/os-labs/week1/linux_lab/** command was used. The output showed the list of all files inside linux_lab. It also showed each file's permissions, ownership, human-readable size, and last-modified timestamp.

View the Content of C File

```
wagleayden@Ayden:~$ cat ~/os-labs/week1/linux_lab/hello.c
#include <stdio.h>

int main(){
    int a = 100, b = 200;

    printf("Hello, Linux Lab!\n");
    printf("Sum of %d and %d is %d.\n", a, b, a + b);
    return 0;
}
wagleayden@Ayden:~$
```

Figure 21: Final Content Check of hello.c

To display the full contents of the source file, **cat ~/os-labs/week1/linux_lab/hello.c** was used. This showed the c file and the txt file together.

Remove the Entire Lab Directory

```
wagleayden@Ayden:~$ sudo rm -rf ~/os-labs/week1/linux_lab
[sudo: authenticate] Password:
wagleayden@Ayden:~$ ls -lh ~/os-labs/week1/
total 20K
-rwxr-xr-x 1 wagleayden wagleayden 16K Sep  6 15:05 practiceayd
-rw-r--r-- 1 wagleayden wagleayden 212 Sep  6 15:04 practiceayd.c
wagleayden@Ayden:~$
```

Figure 22: Removing the linux_lab Directory

To permanently delete the linux_lab directory and all of its contents, **sudo rm -rf ~/os-labs/week1/linux_lab** command was used. It asked for confirmation and afterwards, the contents were successfully removed.

